Health Data Project 1

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Overarching Question

- How do age, gender and occupation affect stress levels and overall sleep, and how do these factors influence heart rate and blood pressure

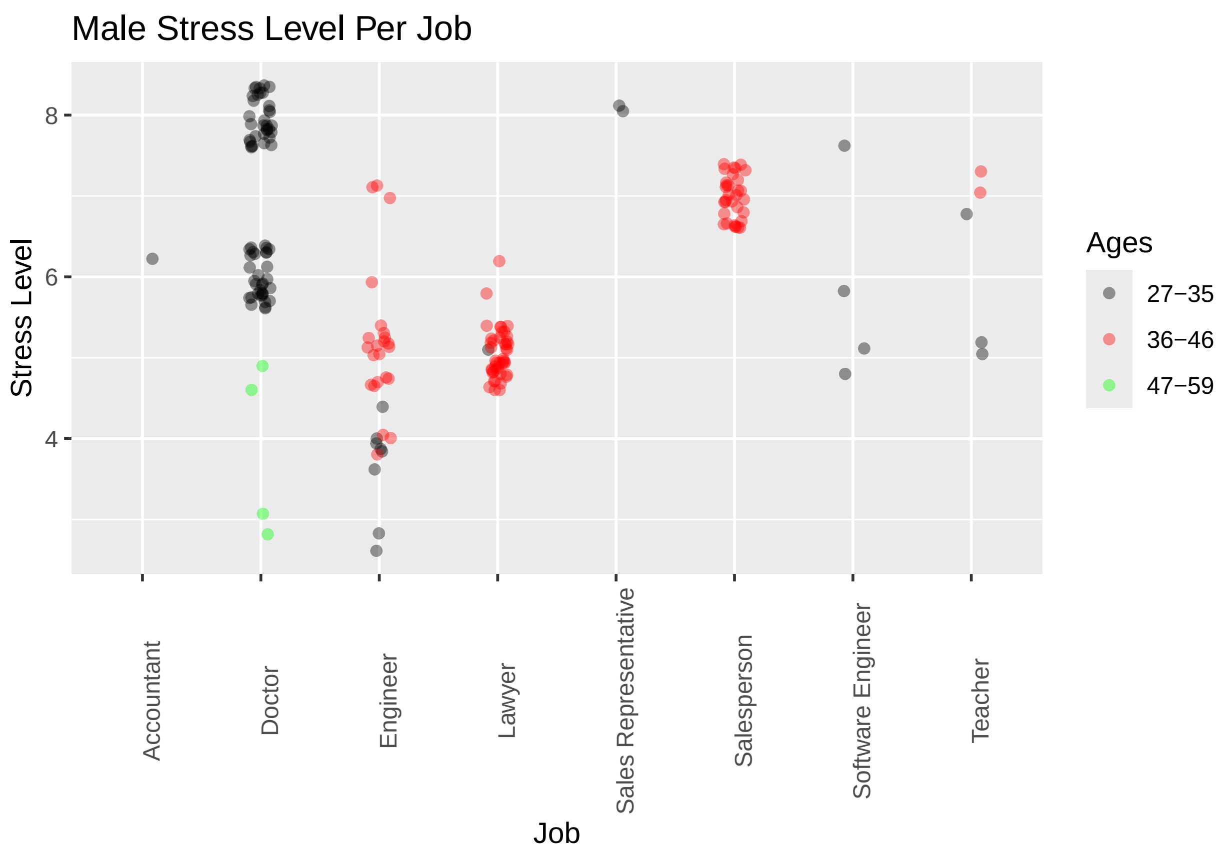
```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3
```

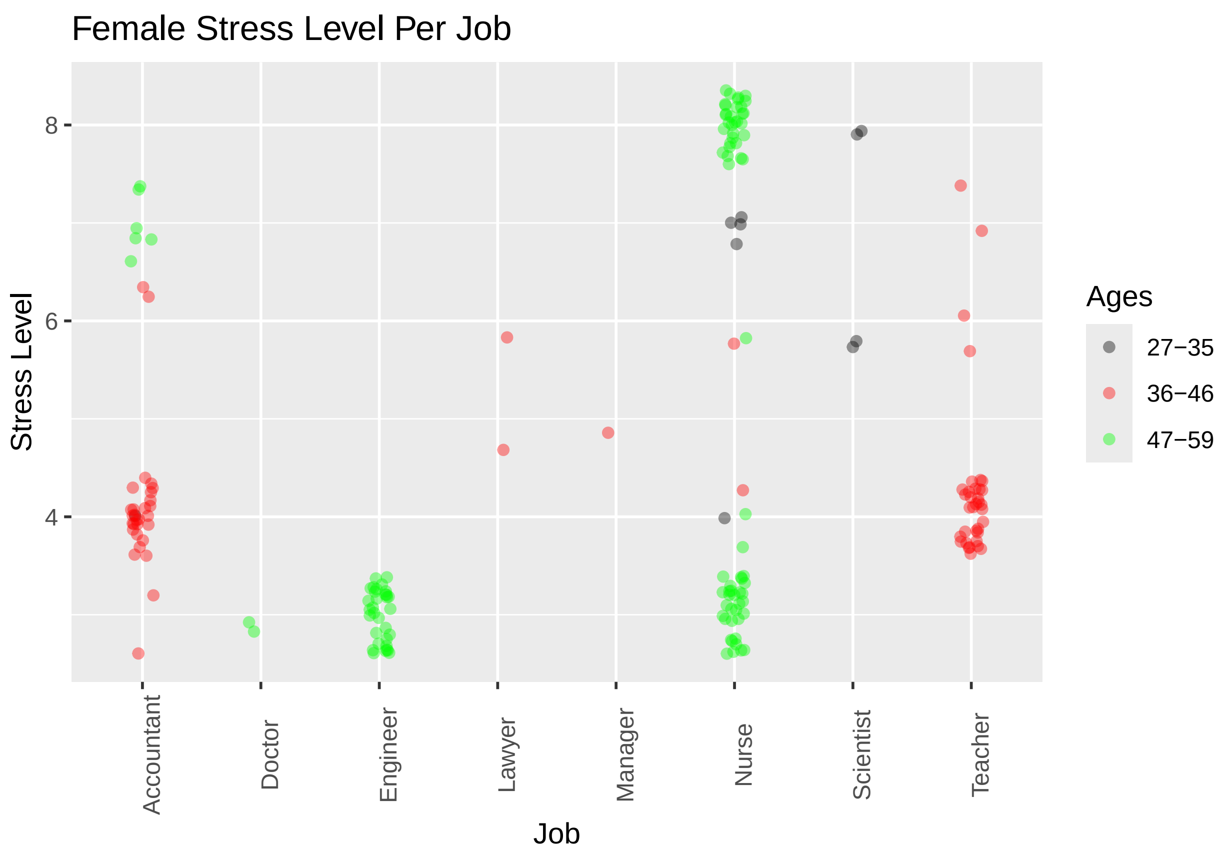
```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.3.3

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
data <- data %>%
  mutate(Ages = case_when(Age >= 27 & Age <= 35 ~ "27-35", Age >= 36 & Age <= 46 ~ "36-46", Age >= 47 &

male_plot <- ggplot(data = filter(data, Gender == "Male"), aes(x = Occupation, y = Stress.Level, color =
  geom_jitter(width = 0.1, alpha = 0.4) +
  labs(title = "Male Stress Level Per Job",x = "Job", y = "Stress Level") +theme(axis.text.x = element_

female_plot <- ggplot(data = filter(data, Gender == "Female"),aes(x = Occupation, y = Stress.Level, colo
  geom_jitter(width = 0.1, alpha = 0.4) +
  labs(title = "Female Stress Level Per Job",x = "Job", y = "Stress Level") +theme(axis.text.x = elemen


print(male_plot)
```




```
print(female_plot)
```



These graphs are sepereated by gender to se if there is a noticable difference between gender in stress levels per occupation. Furthermore the data points in this graph are color coded by age group to see if there is a difference in stress per age as well. The graph for Males is showing how jobs associated with harder workloads like doctors induce a higher stress level. Also interesting to note how for some jobs like like doctors and engineers there is a clear difference in stress levels based on age. For females it is interesting to note the variance in stress levels for women of the same age in the nurse occupation, or for accountants older women experience a higher stress level. Overall we can see that younger age range regardles of gender has a higher variation.

```
data <- data %>%
  mutate(Age_Range = case_when(Age >= 27 & Age <= 35 ~ '27-35',Age >= 36 & Age<= 46 ~ '36-46',Age >= 47

average_stress <- data %>%
  filter(!is.na(Age_Range)) %>%
  group_by(Gender, Age_Range, Occupation) %>%
  summarize(Average_Stress_Level = mean(Stress.Level, na.rm = TRUE)) %>%
  arrange(Gender, Age_Range, Occupation)
```

```
## `summarise()` has grouped output by 'Gender', 'Age_Range'. You can override
## using the `.groups` argument.
```

```
print(average_stress)
```

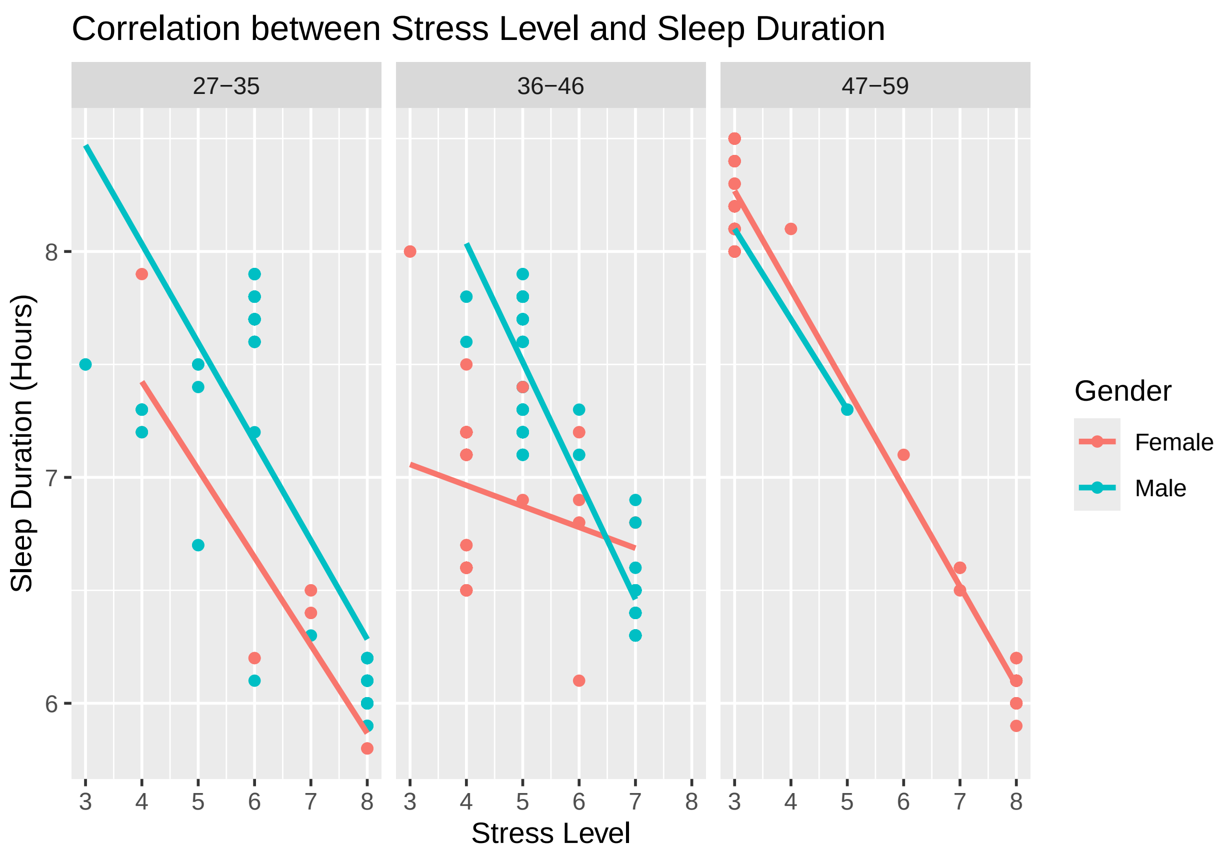
```
## # A tibble: 24 x 4
## # Groups:   Gender, Age_Range [6]
##    Gender Age_Range Occupation Average_Stress_Level
##    <chr>  <chr>     <chr>                     <dbl>
##  1 Female 27-35     Accountant                 4
```

```
##   2 Female 27-35     Nurse                          6.4
##   3 Female 27-35     Scientist                      7
##   4 Female 36-46     Accountant                     4.07
##   5 Female 36-46     Lawyer                         5.5
##   6 Female 36-46     Manager                        5
##   7 Female 36-46     Nurse                          5
##   8 Female 36-46     Teacher                        4.29
##   9 Female 47-59     Accountant                     7
## 10 Female 47-59     Doctor                         3
## # i 14 more rows
```

This shows the averages of stress levels per occupation seperated by gender and age range. This helps us get a good statistical representation of the previous graph. Lowest average stress are female doctors and engineers aged 47-59 with an average stress level of 3. the highest being male sale representatives aged 27-35 with the average stress rating of 8.

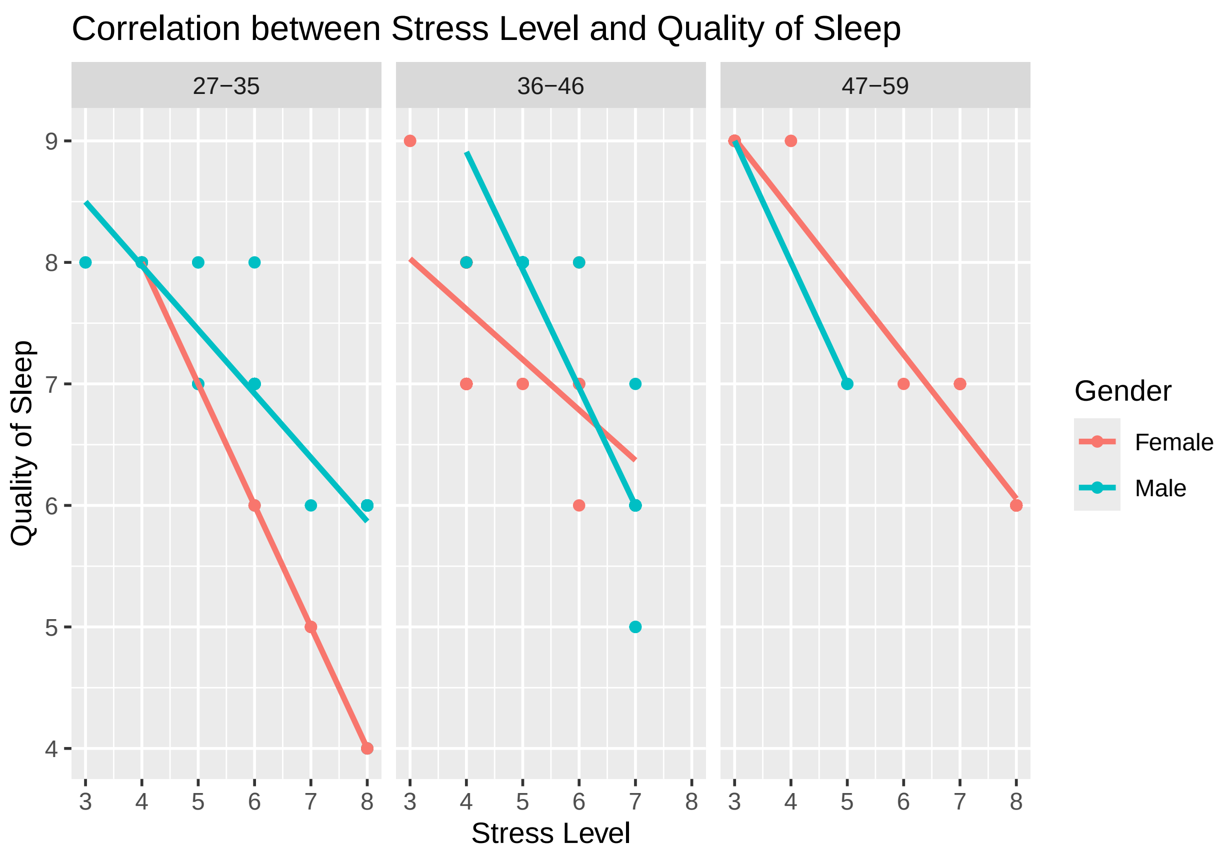
```
data <- data %>%
  mutate(Age_Group = case_when(Age >= 27 & Age <= 35 ~ "27-35",Age >= 36 & Age <= 46 ~ "36-46",Age >= 47


ggplot(data, aes(x = Stress.Level, y = Sleep.Duration, color = Gender)) +
  geom_point(alpha = 1) +
  geom_smooth(method = "lm", se = FALSE) +
  facet_wrap(~ Age_Group) +
  labs(title = "Correlation between Stress Level and Sleep Duration",x = "Stress Level", y = "Sleep Dura
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
ggplot(data, aes(x = Stress.Level, y = Quality.of.Sleep, color = Gender)) +
  geom_point(alpha = 1) +
  geom_smooth(method="lm", se = FALSE) +
  facet_wrap(~ Age_Group) +
  labs(title = "Correlation between Stress Level and Quality of Sleep", x = "Stress Level",y = "Quality
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



For sleep duration we see that there is a slightly sharper decline for women ages 27-35, however in the duration for men and women aged 36-46 we see that men maintain that same general trend, whereas women face a significnly more gradual decline. For the 47-59 age group women face a slight sharper decline. For quality of sleep women aged 27-35 face a sharp decline same in ages from 36-46, males have a sharper decline in ages 47-59. Some things to takeaway are that there are sharper declines in older individuals for duration of sleep when dealing with higher stress. However in terms of quality of sleep despite staring off with a higher quality as the stress increases the decline for quality of sleep as stress increases is sharper. Overall both duration and quality show to decrease as stress increases however quality of sleep is much more related to stress, showing that the quality decreases even if there is still a high amount of sleep, showing that the affect of stress is greater on quality than quantity.

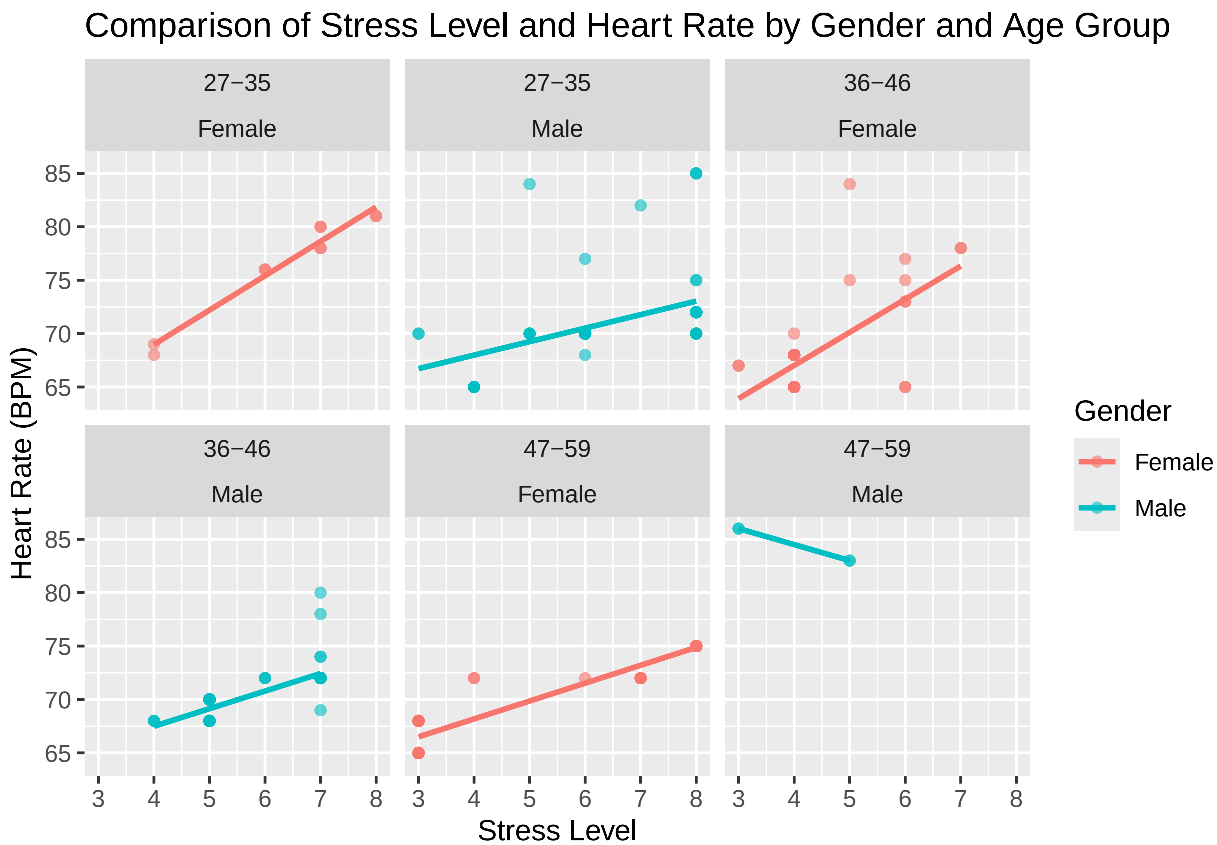
```
library(ggplot2)
library(dplyr)
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 4.3.3
```

```
data <- data %>%
  separate(Blood.Pressure, into = c("Systolic_BP", "Diastolic_BP"), sep = "/", convert = TRUE)

data <- data %>%
  mutate(Age_Group = case_when(Age >= 27 & Age <= 35 ~ "27-35",Age >= 36 & Age <= 46 ~ "36-46", Age >= 

data_filtered <- data %>%
  filter(Age_Group != "Other")


ggplot(data_filtered, aes(x = Stress.Level, y = Heart.Rate, color = Gender)) +
```

```
  geom_point(alpha = 0.6) +geom_smooth(method = "lm", se = FALSE) +facet_wrap(~ Age_Group + Gender) +
  labs(title = "Comparison of Stress Level and Heart Rate by Gender and Age Group",x = "Stress Level",y
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

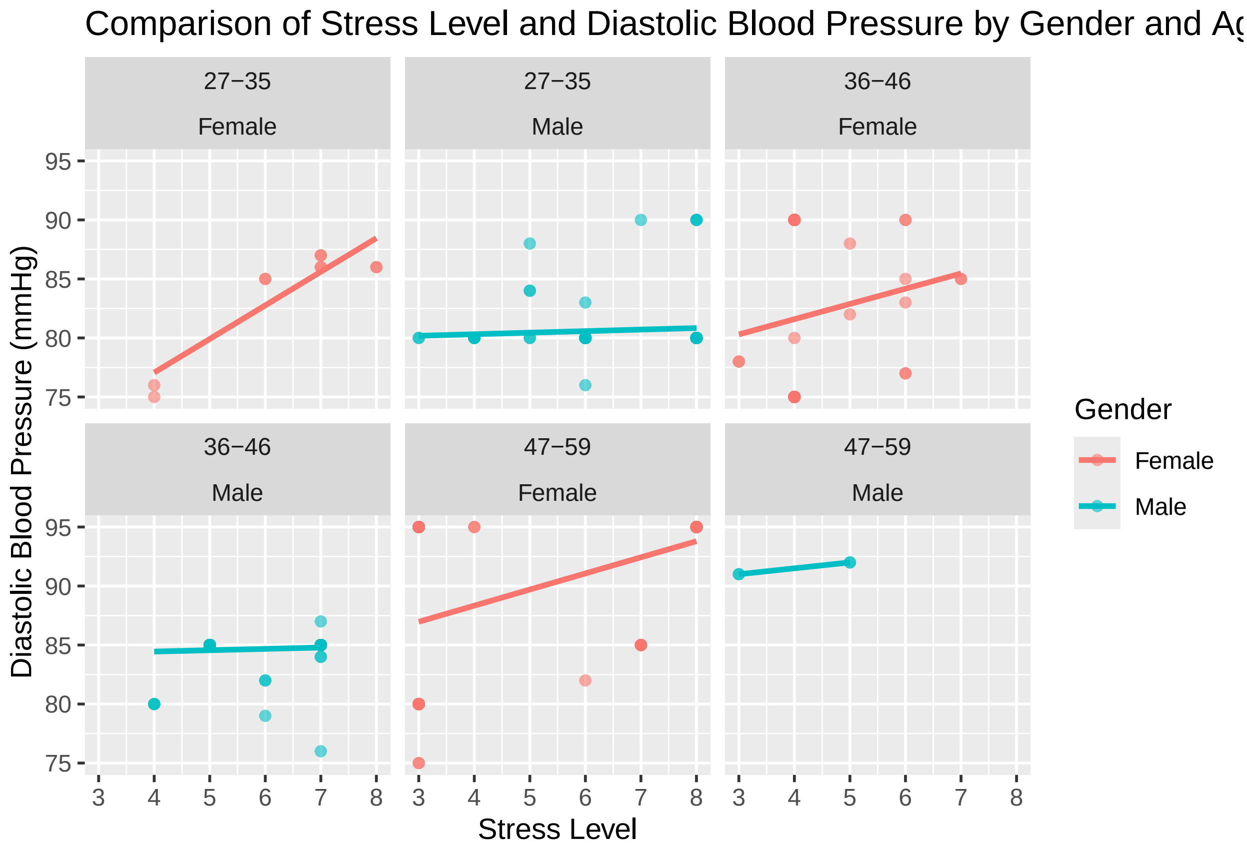



```
ggplot(data_filtered, aes(x = Stress.Level, y = Systolic_BP, color = Gender)) +
  geom_point(alpha = 0.6) + geom_smooth(method = "lm", se = FALSE) + facet_wrap(~ Age_Group + Gender) +
  labs(title = "Comparison of Stress Level and Systolic Blood Pressure by Gender and Age Group", x = "S
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

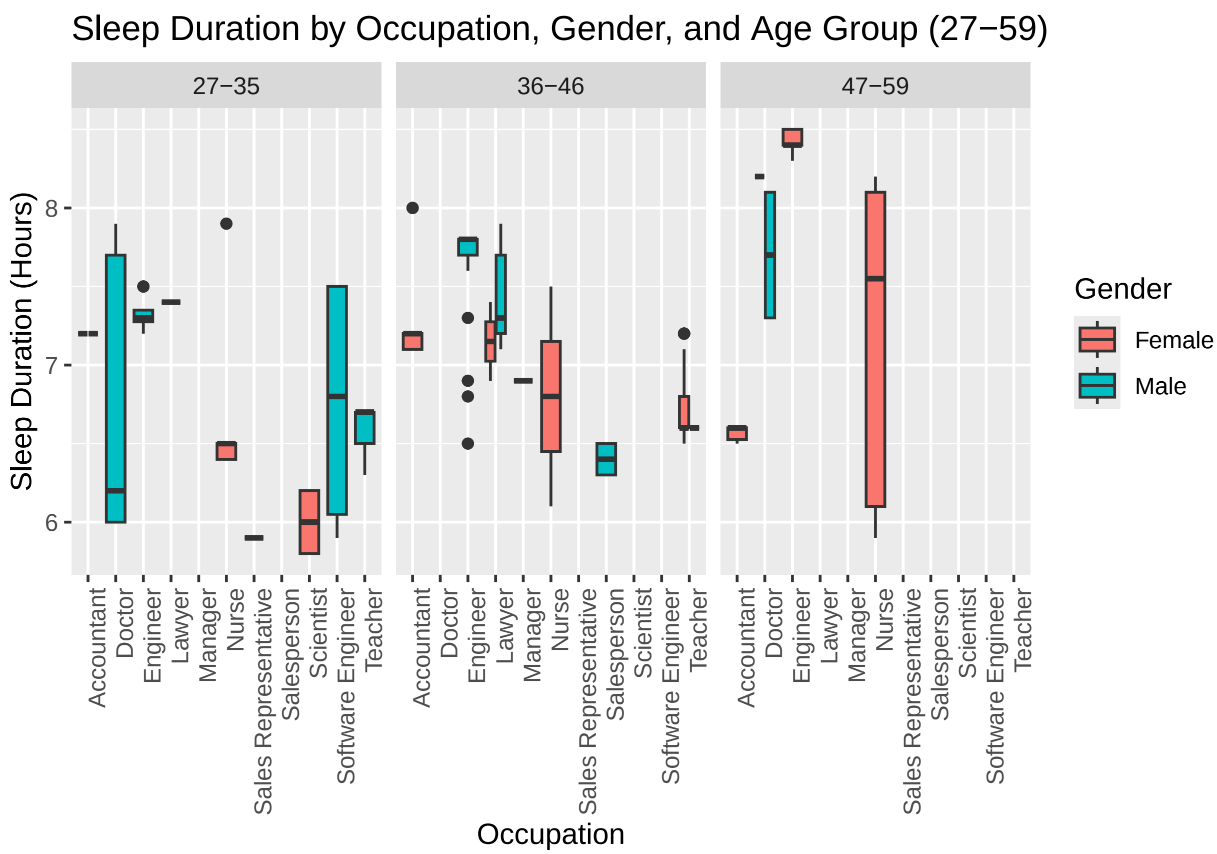

```
ggplot(data_filtered, aes(x = Stress.Level, y = Diastolic_BP, color = Gender)) +
  geom_point(alpha = 0.6) + geom_smooth(method = "lm", se = FALSE) + facet_wrap(~ Age_Group + Gender) +
  labs(title = "Comparison of Stress Level and Diastolic Blood Pressure by Gender and Age Group", x = "S
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

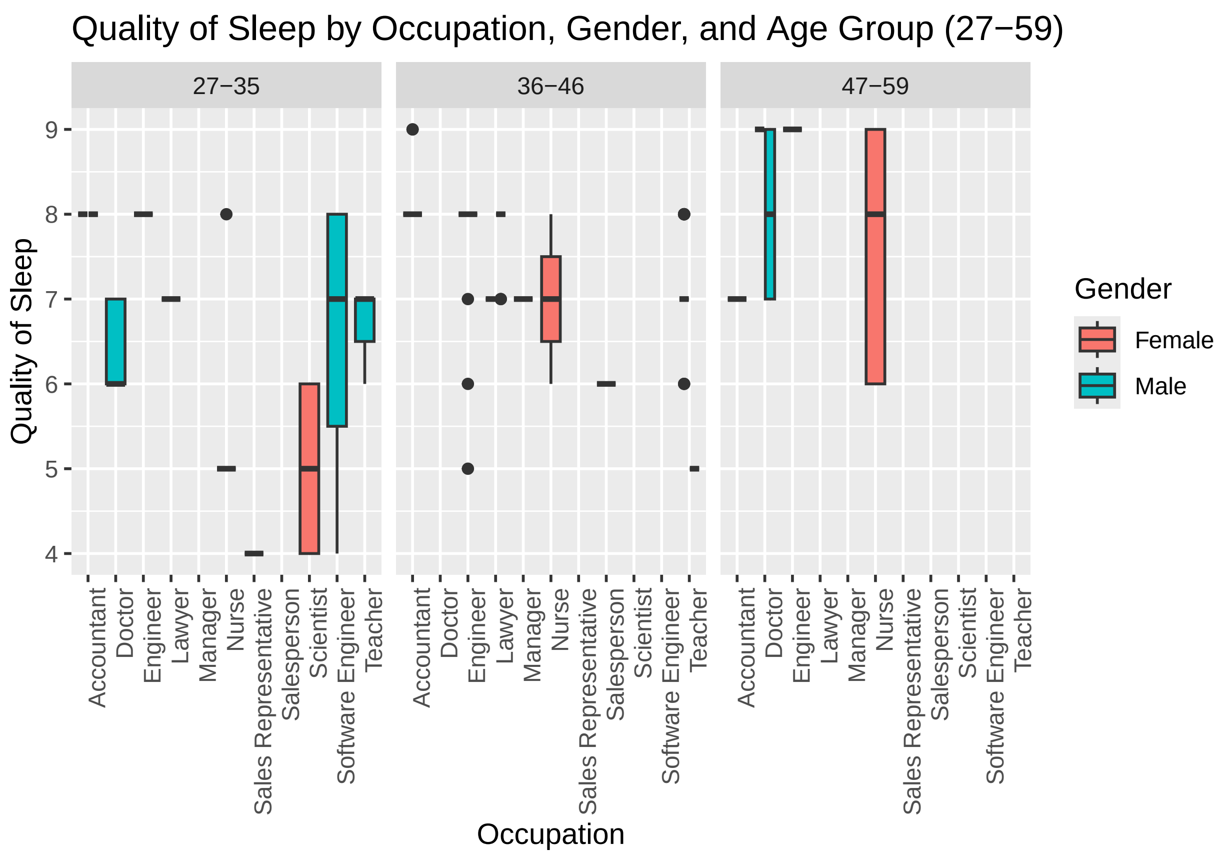


This graph is showing the relation between stress and both heart rate and blood ressure. The original dataset had blood ressure combined, so I had to to split it in systolic and diastolic in order for the data to be graphed properly. These graphs are all generally the same with the exception of heart rate for men aged 47-59. However the main takeaway for these graphs is that women are more privy to increased heart rate and blood pressure as stress levels increase, this is the same for women across all age groups. Another iportant aspect is that the increase in heart rate and blood pressre in relation to stress increase relativley the same for women whereas for men you notice a slight increase in those aspects as they get older.

```
data <- data %>%
  mutate(Age_Group = case_when(Age >= 27 & Age <= 35 ~ "27-35",Age >= 36 & Age <= 46 ~ "36-46",Age >= 4
  filter(!is.na(Age_Group))


ggplot(data, aes(x = Occupation, y = Sleep.Duration, fill = Gender)) +
  geom_boxplot() +
  facet_wrap(~Age_Group) +
  labs(title = "Sleep Duration by Occupation, Gender, and Age Group (27-59)", y = "Sleep Duration (Hour
```




```r
ggplot(data, aes(x = Occupation, y = Quality.of.Sleep, fill = Gender)) +
  geom_boxplot() +
  facet_wrap(~Age_Group) + labs(title = "Quality of Sleep by Occupation, Gender, and Age Group (27-59)"
```



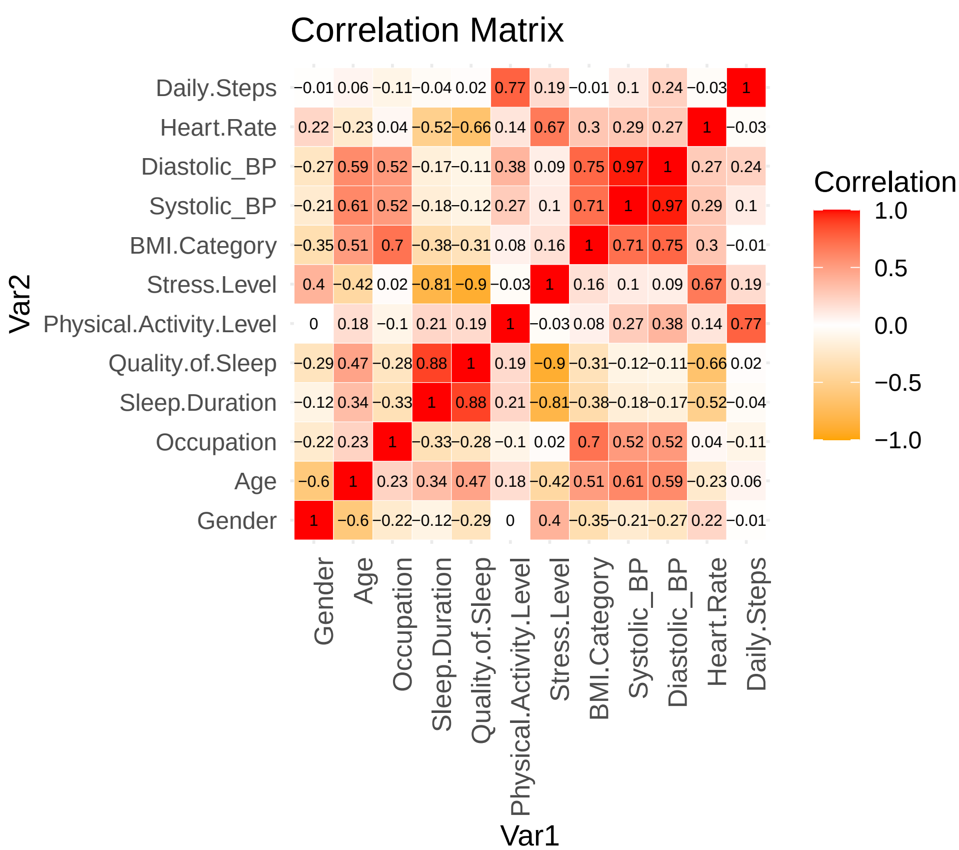
This graph is showing the effects of occupation on quality and quantity of sleep. In many occupations women have better sleep quality than men even when they are getting simmilar amount of sleep, furthermore this graph shows that 47-59 year olds have more consistent overall sleep. Most importantly this graph shows that jobs that typically are assosciated with higher stress like doctors and scientists, and managers have both lower sleep quality and quantity.

```
library(dplyr)
library(ggplot2)
library(reshape2)
```

```
## Warning: package 'reshape2' was built under R version 4.3.3

##
## Attaching package: 'reshape2'

## The following object is masked from 'package:tidyr':
##
##     smiths
```

```
if ("BMI.Category" %in% colnames(data) && (is.character(data$BMI.Category) || is.factor(data$BMI.Catego
  data$BMI.Category <- as.numeric(factor(data$BMI.Category))
}

if ("Occupation" %in% colnames(data)) {
  data$Occupation <- as.numeric(factor(data$Occupation))
}

if ("Gender" %in% colnames(data)) {
  data$Gender <- as.numeric(factor(data$Gender))
```

```
}

numeric_data <- data %>%
  select(-Person.ID, -Sleep.Disorder) %>%
  select_if(is.numeric)

correlation_matrix <- cor(numeric_data, use = "complete.obs")

correlation_melted <- melt(correlation_matrix)

ggplot(data = correlation_melted, aes(x = Var1, y = Var2, fill = value)) + geom_tile(color = "white") +
  geom_text(aes(label = round(value, 2)), color = "black", size = 2) +
  theme_minimal() +theme(axis.text.x = element_text(angle = 90, vjust = 1, size = 10, hjust = 1)) +coor
```



This is a correlation matrix showing the corelation between all values in the dataset. Age, gender and occupation do have an influence on stress levels which ultimatley indeed affect sleep, heart rate and blood pressure. However This is not as strong as a link as previously thought factors like Age plays a stronger role in blood pressure than stress. Gender plays a role in bo th stress and heart rate. Overall we can see the importance of stress management and the dtermintal effects it can have on your health

Ultimatley based on the graphs and correlation matrix we do in fact see that there is a relation between stress levels and age, gender and occupation, and consequently we see these effects of overall sleep and cardiovascular health, however the relation between these factors is not as strong as I had previosuly thought before analyszing the data. While the relation is there other factors like age, and gender that effect sleep and cardiovascular health more than the actual stress does.